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Control strategy of regenerative braking system in electric vehicles

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Abstract

Regenerative braking technique, which can improve the efficiency of energy conversion and increase the driving range, is one of the key technologies of core competitiveness of electric vehicles (EVs). In this paper, a novel control strategy of regenerative braking is proposed based on a novel definition method of braking intensity, which is applied to all braking condition. Then, according to the slope information, the dynamic distribution algorithm of the front and rear axle is presented. Next, based on logic threshold control algorithm with the input of z and I, the distribution relationship between electric and mechanical braking is determined. Finally, the effectiveness of the proposed control strategy has been validated in MATLAB/Simulink. Simulation results show that the modified control strategy is effective on road conditions with different slope.

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Keyword: electric vehicle, regenerative braking, the down slope driving, braking stability;

1. Introduction

With the increasingly deterioration of environment pollution and the growing shortage of fuel fossils, electric vehicles (EVs) are undertaking heavy responsibilities to ease the problem of the harsh environment and energy depletion in recent years [1]. Power management strategy [2-5] is a key technology to improve the EV performances. Regenerative braking, which is the process of energy conversion, refers to transform the kinetic energy and potential

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energy of the braking and declivity into electric energy by generator and restore it into the energy storage devices.

Some literatures have reported research on the regenerative braking control strategy, most of which are confined to flat driving cycle. Nevertheless, domestic and foreign scholars have little research on the regenerative braking control of downhill. Three typical braking distribution strategies are proposed for parallel hybrid vehicles, namely parallel, optimal sensory series and optimal energy recovery series braking strategy [6]. With the future information of vehicle operation status from GPS, Shu etc. established the local optimization algorithm in predict vision to realize the rolling optimization control with model prediction method [7]. Han etc. put forward the control method of feed-forward and feedback to coordinate the motor and hydraulic auxiliary brake system, and effectively guaranteed the response and precision of the auxiliary system [8]. In order to improve the efficiency of energy recovery and take into account the braking safety and stability simultaneously, it is necessary to put forward a regenerative braking strategy considering the down slope driving and braking stability [9].

In view of the shortcomings of existing researches, the specific research contents of this paper are as follows: (1) the change of braking force distribution and braking stability considering the down slope; (2) the influence of slope on regenerative braking recovery and safety. The remainder of this paper is organized as follows: the vehicle configuration is described in Section 2. The mechanical analysis based on downhill are proposed in Section 3. The regenerative braking control strategy with cruising downhill is expressed in Section 4. Simulation results and analysis are carried out in Section 5, and conclusions are given in Section 6.

Nomenclature

a	the distance from the front axle to the centroid	α	the slope of road
b	the distance from the rear axle to the centroid	m	the vehicle mass
F_{xb}	the required total braking force	g	gravity
F_{xb1}	maximum braking force in front wheels	L	wheelbase
F_{xb2}	maximum braking force in rear wheels	F_{u1}	the brake force of front axle
β	the braking torque distribution coefficient	F_{u2}	the brake force of rear axle
F_{xbmax}	the maximum braking force on the ground	z	braking intensity
φ	adhering coefficient of the road surface	i	the value of slope
F_{z1}	the perpendicular force of the front wheels	k_i	the impact factor of the slope
F_{z2}	the perpendicular force of the rear wheels	h_g	the height of centroid

2. The vehicle configuration

A single-shaft electric vehicle driven by front axle is used in this paper, which configuration is shown in Fig. 1. There is one power sources, namely the electric motor, and two part of brake system. One part corresponds to the pneumatic braking system. The other part consists of electric braking system, which is mainly composed of battery, motor and transmission system. The management level consists of electronic controller unit, battery management system, motor controller and brake controller, which communicate via the CAN protocol. Under the controllers' cooperative control, the hydraulic actuator and the generator realize the baking process as the actuator of mechanical braking and electric braking apart.

3. Basic theoretical analysis based on downhill braking

3.1 Vehicle dynamic model

According to the structure of the vehicle drive system described in Section 2, the vehicle dynamics model in braking process of downhill is built considering load transfer, as shown in Fig.2. The inertia force moment of rolling resistance, air resistance and rotation mass is ignored. In addition, neglecting the braking process of rolling and sliding, we think that the adhesion coefficient is only a definite value in the following analysis. According the vehicle dynamics equations, the horizontal force equations along the ramp are given by:

$$F_{xb} = F_{xb1} + F_{xb2} = mg\left(z_a + \sin\alpha\right) \tag{1}$$

When the wheels are locked, the maximum braking force on the ground is expressed as:

$$F_{xb\max} = \varphi mg\cos\alpha \tag{2}$$

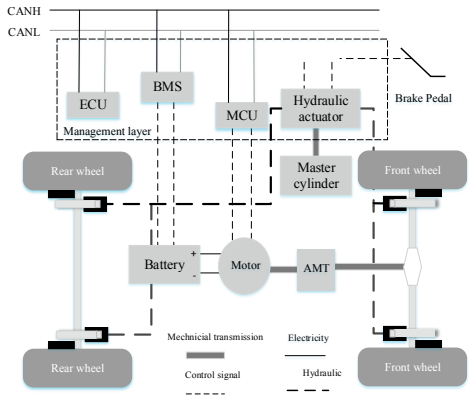


Fig. 1. The configuration of RBS.

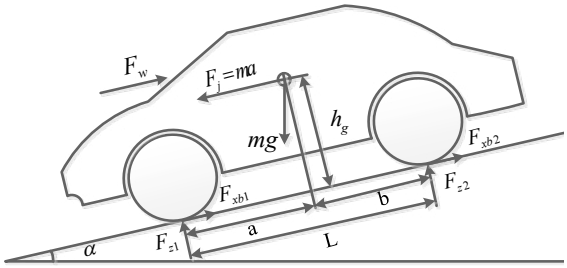


Fig. 2. Kinetic analysis of vehicle in braking process.

3.2 A novel definition of braking intensity

Compared to the process of brake on the horizontal road, the driver should increase the brake pedal force or start the auxiliary braking cruise device up to keep steady running on the slope road, and then prevent the traffic accidents caused by rear-end collision. In general, braking intensity is defined as $z = a/g$. However, it is not suitable for cruising downhill regenerative braking (CDRB). To reduce the computational complexity of the braking process and response the integrity of the braking intensity definition, we redefine the braking intensity as $z = a/g + \sin\alpha$. When and only when the wheel is locked, the braking intensity is equal to $\varphi\cos\alpha$.

3.3 The change of brake control line based on downhill braking

The front and rear wheels are simultaneously locked in the process of braking, which is beneficial to the application of adhesion condition and the directional stability of braking. The distribution relationship of braking force on the road of any friction coefficient is written as:

$$\begin{cases} F_{u1} + F_{u2} = \varphi mg\cos\alpha \\ F_{u1} = \varphi F_{z1} = \dfrac{G\left(b\cos\alpha + zh_g\right)}{L} \\ F_{u2} = \varphi F_{z2} = \dfrac{G\left(\mathrm{a}\cos\alpha - zh_g\right)}{L} \end{cases} \tag{3}$$

$$F_{u2}=\frac{1}{2}\left[\frac{G\cos\alpha}{h_g}\sqrt{b^2+\frac{4h_gL}{G\cos\alpha}F_{u1}}-\left(\frac{Gb\cos\alpha}{h_g}+2F_{u1}\right)\right] \tag{4}$$

Because of the motor placed in the front wheel, the front wheels are utilized to realize energy recovery during braking. In order to prevent premature lock of the front wheels due to excessive braking, it is necessary for the front wheels to restrain in the regenerative control strategy. On the basic of the equation (3) and (4), f and r line groups are expressed as:

$$\begin{cases}F_{xb2}=\dfrac{L-\varphi h_g}{\varphi h_g}F_{xb1}-\dfrac{Gb\cos\alpha}{h_g}\\ F_{xb2}=\dfrac{-\varphi h_g}{L+\varphi h_g}F_{xb1}+\dfrac{\varphi Ga\cos\alpha}{L+\varphi h_g}\end{cases} \tag{5}$$

4. Control strategy of regenerative braking

4.1 The principle of upper braking force distribution

In order to ensure the directional stability and enough efficiency in the process of braking, ECE R13 braking regulations constituted by the United Nations economic commission for Europe put forward explicit requirements for front and rear braking force of dual-shaft vehicle. Under various loading conditions, we must ensure that the braking strength is satisfied; we apply the new definition of the braking intensity proposed in Section 3 to explore the ECE R13 braking regulation. In the light of the definition of adhesion coefficient, the value range of dynamic distribution coefficient can be obtained as follow:

$$\begin{cases}\beta\le\dfrac{(z+0.07)}{0.85}\times\dfrac{b\cos\alpha+zh_g}{zL} & (0.1\le z\le 0.61)\\ \beta\le 1-\dfrac{(z+0.07)}{0.85}\times\dfrac{a\cos\alpha-zh_g}{zL} & (0.1\le z\le 0.61)\\ \beta>\dfrac{b\cos\alpha+zh_g}{L\cos\alpha} & (0.1\le z\le 0.61)\end{cases} \tag{6}$$

As shown in Fig. 3, when the braking intensity is less than 0.1, the distribution coefficient of braking force can obtain any value between 0 and 1. With the increase of the pavement slope, when the braking intensity is between 0.1 and 0.61, the range of the braking distribution coefficient is gradually narrowed. Therefore, we need to strictly control the braking distribution coefficient within a reasonable range to ensure the braking safety and the efficiency of energy recovery. Because the front wheels, which are played by the motor, are adopted to recover energy, braking force distribution coefficient would be carried out in accordance with the higher limit of braking regulations, braking force distribution in front and rear wheels is shown in Fig. 3. The braking distribution relationship of the front and rear wheels according to ECE regulations is shown as the following formula:

$$F_{u2}=\frac{1}{2}\left[\frac{G}{h_g}\sqrt{\left(b\cos\alpha+0.07h_g\right)^2-4h_g\left(0.07b\cos\alpha-\frac{0.85L}{G}F_{u1}\right)}-\left(\frac{G}{h_g}\left(b\cos\alpha+0.07h_g\right)+2F_{u1}\right)\right] \tag{7}$$

According to the above formula (4), (5) and (8), we can obtain the maximum dynamic distribution function of the front braking force on braking intensity. According to the above study, the distribution relationship of the front and

rear wheels is shown in Fig. 4. Due to the difference of the braking control lines between the flat road and slope road, A, B and C are dynamically moved to A', B' and C'. Therefore, the dynamic threshold value of the top-level braking force distribution needs to be transformed in the BCU controller according to the gradient information returned in vehicle-mounted GPS or main parameters of vehicle estimator to ensure the braking security.

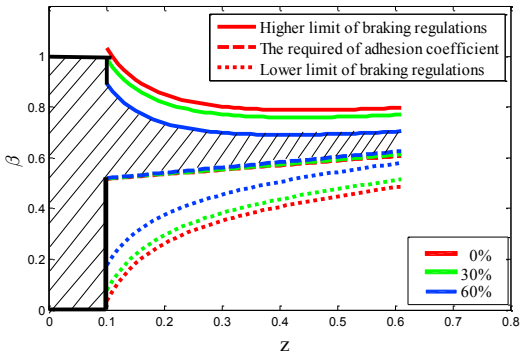


Fig. 3. The range of braking distribution coefficients

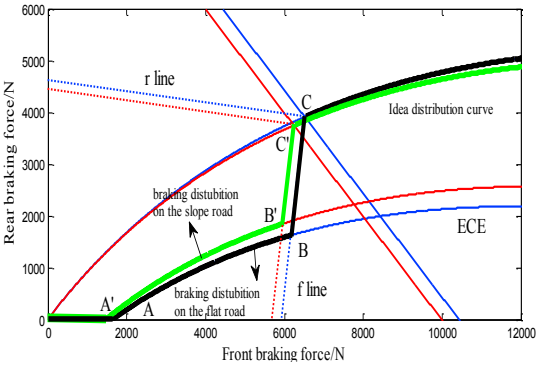


Fig. 4. The principle of braking distribution

4.2 The control strategy of bottom distribution based on logic threshold

The application of novel definition method can extend the adaptive scope of conventional regenerative braking control strategy and recover more energy. The variation of braking intensity is only related to slope when the vehicle is on the cruise downhill. According to the maximum longitudinal gradient of highway grade, the range of slope is 3~9%. The slope impact factor to explain the influence of slope information on RB, expressed by

$$k_i = \begin{cases} 1 & 0 \le i \le 15\% \\ \dfrac{i-15}{15} & 15\% \le i \le 30\% \\ 0 & 30\% \le i \le 100\% \end{cases} \tag{8}$$

The electric braking torque generated by the motor is applied to the front wheels. When z is less than 0.1, the braking system enters the pure electric braking mode. When z is between 0.1 and 0.7, the braking system enters the composite braking mode. However, if the total required torque cannot be satisfied by regenerative braking torque, the motor would provide its max regenerative torque, and the remaining torque would be compensated by the mechanical braking system. When z is more than 0.7, RBS shifts to the mechanical braking mode to ensure the braking safety.

5. Simulation result and analysis

In order to verify the proposed control strategy, we conduct simulation and analysis in MATLAB/Simulink software environment. The effectiveness of the control strategy is verified. The simulation result is shown in Fig. 5. In the 15%~30% road surface defined as a steep slope road, the modified control strategy is more smoothly switched between electric braking and mechanical braking, preventing the frustration and instability caused by switching of braking mode.

6. Conclusion

In this paper, the control strategy of regenerative braking considering down slope is proposed on electric vehicles. The proposed algorithm is verified by the simulation of driving cycle. The energy recovery of regenerative braking

can be successfully carried out in the down slope driving. The results show that the regenerative braking control strategy can effectively recover energy and improve the braking stability.

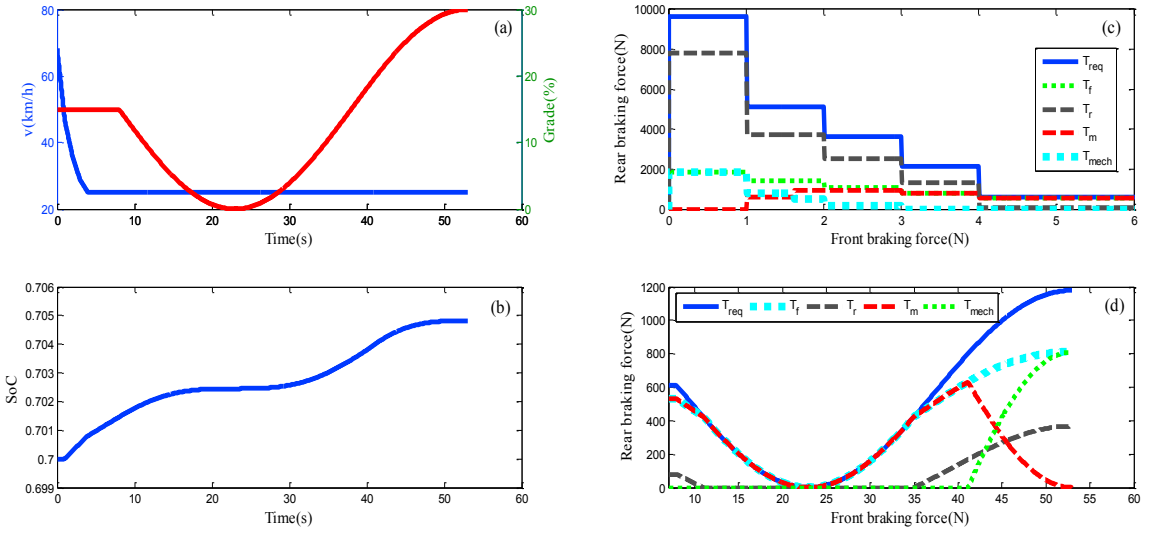


Fig. 5. The custom driving cycle (a) the velocity of the driving cycle and the grade of road (b) the change of SOC; Simulation result (c) the relationship of the various braking force in 0~6s; (d) the relationship of the various braking force in 7~60s.

Acknowledgements

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